Table 1. Loosening Torque Measurements

| Attachment Bolt Identification | Torque (pound-force feet) | Attachment Bolt Identification | Torque (pound-force feet) |
|---|---|---|---|
| **RA-1** | 11.9 | **RC-1** | <5.0 |
| **RA-2** | 17.0 | **RC-2** | 19.0 |
| **RA-3** | 22.4 | **RC-3** | 23.9 |
| **RA-4** | 19.7* | **RC-4** | 29.2 |
| **RB-1** | 10.0 | **RC-5** | <5.0 |
| **RB-2** | 21.0 | **RD-1** | <5.0 |
| **RB-3** | 17.8 | **RD-2** | 9.5 |
| **RB-4** | 17.7 | **RD-3** | 10.7 |
| **LA-1** | 5.1 | **RD-4** | 10.9 |
| **LA-2** | 11.8 | **RD-5** | 5.8 |
| **LA-3** | 14.6 | **LC-1** | 8.3 |
| **LA-4** | 9.9 | **LC-2** | 19.5 |
| **LB-1** | 8.3 | **LC-3** | 21.0 |
| **LB-2** | 20.4 | **LC-4** | 21.9 |
| **LB-3** | 26.0 | **LC-5** | 10.0 |
| **LB-4** | 23.8 | **LD-1** | 9.3 |
| | | **LD-2** | 17.9 |
| | | **LD-3** | 19.7 |
| | | **LD-4** | 23.5 |
| | | **LD-5** | 8.9 |

*After movement had been detected with tightening torque applied.

A piece of the main spar cap from the left wing was sent to Lehigh Testing Laboratories, Inc., New Castle, Delaware, for tension testing and chemical analysis. The sent piece was an approximately 10.5 inches long section of the lower spar cap with the web cut nearly flush to the upper surface of the spar cap. To prepare the piece for shipment to the test lab, rivets attaching the wing lower skin were drilled out to separate the skin from the spar cap. Then the piece was soaked in paint stripper to chemically remove the paint in preparation for machining of tensile specimens.

At Lehigh Testing, four subsize tensile specimens were machined from the lower spar cap from the middle portion between the rivet holes and outboard attachment bolt holes. Additional machining was required on the upper face to reduce the tensile specimen thickness to within tension specimen standard dimensions in accordance with ASTM Standard E8/E8M.[3] The lower surface remained in the as-received condition on the tensile specimens.

[3] ASTM E8/E8M – 16a, *Standard Test Methods for Tension Testing of Metallic Materials*, ASTM International (2016).